Exposé VI. The $K^\bullet$ of a projective bundle: computations and consequences

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1. Computation of $K^\bullet$ of a projective bundle: the case of locally free sheaves of finite type

In this section $K^\bullet(X)$ denotes the Grothendieck group of locally free sheaves of finite type on a scheme X. The aim is to prove the following structure theorem.

Theorem 1.1. Let S be a quasi-compact scheme, let E be a locally free $\mathcal{O}_S$-module of rank $r+1$, and let

$$X = \mathbf{P}(E)$$

be the associated projective bundle. The structural morphism $f : X \to S$ gives $K^\bullet(X)$ the structure of a $K^\bullet(S)$-algebra. If $\xi \in K^\bullet(X)$ denotes the class of $\mathcal{O}_X(1)$, then $K^\bullet(X)$ is a free $K^\bullet(S)$-module with basis

$$1, \xi, \ldots, \xi^r.$$

1.2. Plan of the proof

The proof is divided into three parts.

First, one proves that the powers of ξ generate $K^\bullet(X)$ as a $K^\bullet(S)$-module. For this it is enough to establish the following facts. If F is locally free of finite type on X, one can find a graded $\mathrm{Sym}(E)$-module M, of finite presentation and flat over $\mathcal{O}_S$, such that $F = \mathrm{Proj}_0(M)$. Such an M admits a finite graded resolution

$$0 \longrightarrow L_{r+1} \longrightarrow \cdots \longrightarrow L_0 \longrightarrow M \longrightarrow 0$$

by graded $\mathrm{Sym}(E)$-modules locally free of finite type. Finally, each L_i has a finite filtration whose quotients are of the form

$$N \otimes_{\mathcal{O}_S} \mathrm{Sym}(E)(n),$$

with N locally free of finite type on S. Applying Proj_0 then gives the required generation by the powers of ξ.

Second, one proves that $1, \xi, \ldots, \xi^{r+1}$ are linearly dependent over $K^\bullet(S)$. This is obtained from the canonical epimorphism $f^*E \to \mathcal{O}_X(1)$.

Third, one proves that $1, \xi, \ldots, \xi^r$ are linearly independent over $K^\bullet(S)$.

Let S be a scheme, let $B = \bigoplus_{n\geq 0} B_n$ be a graded $\mathcal{O}_S$-algebra, and let M be a graded B-module. If N is a graded B-module, $N(n)$ denotes the graded module defined by

$$N(n)_k = N_{n+k}.$$

A graded B-module is said to be free, respectively locally free, respectively locally free of finite type, if locally on S it is a direct sum, respectively locally a direct sum, respectively locally a finite direct sum, of modules $B(n)$. It is said to be of finite presentation if, locally on S, it admits a presentation by free graded modules of finite type.

For every integer n there is a canonical homomorphism of graded B-modules

$$M_n \otimes_{\mathcal{O}_S} B(-n) \longrightarrow M, \tag{1.2.1}$$

defined by $x \otimes a \mapsto ax$. A free graded module of finite type admits a decomposition

$$M \simeq \bigoplus_{n=n_0}^{n_1} B(-n)^{k_n}. \tag{1.2.2}$$

Proposition 1.2. The integers k_n occurring in a decomposition (1.2.2) are independent of the chosen decomposition.

Indeed, after restricting to an open set where $B_0 = \mathcal{O}_S$, one compares the first non-zero homogeneous component. This determines k_{n_0}; passing to the quotient and arguing by induction gives all the k_n.

Proposition 1.3. Let S be quasi-compact, let E be a locally free $\mathcal{O}_S$-module of finite type, let $X = \mathbf{P}(E)$, and let F be an $\mathcal{O}_X$-module of finite presentation which is flat over S. Then there exists an integer n_0 such that, for all $n \geq n_0$,

(i) $R^i f_*(F(n)) = 0 \quad (i \geq 1)$,

(ii) $f_*(F(n))$ is locally free of finite type on S,

(iii) $M = \bigoplus_{n \geq n_0} f_*(F(n))$ is a graded $\operatorname{Sym}(E)$-module of finite presentation.

The statement is local on S. Since S is quasi-compact, one is reduced to the affine case. If S is noetherian, (i) is Serre's theorem, (ii) follows from EGA III, 7.9.10, and (iii) from the usual comparison between a coherent sheaf on a projective bundle and its associated graded module.

In the general affine case one uses passage to the limit. Write $S = \operatorname{Spec}(A)$ as a filtered projective limit of noetherian affine schemes $S_\lambda = \operatorname{Spec}(A_\lambda)$, with affine transition morphisms. The locally free module E, the scheme X, and the sheaf F descend, after increasing λ, to corresponding data $E_\lambda, X_\lambda, F_\lambda$, with F_λ flat over S_λ. The noetherian case applies to F_λ, and the theorem of change of base gives the assertions after pullback to S.

In this reduction one uses the Cartesian square

$$\begin{tikzcd} X \arrow[r, "g"] \arrow[d, "f"'] & X_0 \arrow[d, "f_0"] \\ S \arrow[r, "g_0"'] & S_0. \end{tikzcd}$$

Here S_0 is noetherian, $X_0 = \mathbf{P}(E_0)$, and $F = g^*(F_0)$. The previous paragraph proves the assertion for F_0, and EGA III, 7.7.5 and 7.8.5 give the required base-change identifications for F.

Corollary 1.4. Under the hypotheses of Proposition 1.3 there exists a graded $\operatorname{Sym}(E)$-module M, of finite presentation and flat over $\mathcal{O}_S$, such that

$$F = \operatorname{Proj}_0(M).$$

Proposition 1.5. Let A be a ring, let B be an A-algebra of finite presentation, and let M be a B-module of finite presentation. Suppose that B and M are flat over A, and that, for every residue field k of A, the $B \otimes_A k$-module $M \otimes_A k$ has projective dimension at most n. Then M has a resolution of length n by projective B-modules of finite type:

$$0 \longrightarrow L_n \longrightarrow \cdots \longrightarrow L_0 \longrightarrow M \longrightarrow 0.$$

This is the local criterion of EGA IV, 12.3.2.

Proposition 1.7. Let S be quasi-compact, let E be locally free of rank $r+1$, let $B = \mathrm{Sym}(E)$ with its usual grading, and let M be a graded B-module of finite presentation, flat over S. Then M admits a graded resolution

$$0 \longrightarrow L_{r+1} \longrightarrow \cdots \longrightarrow L_0 \longrightarrow M \longrightarrow 0,$$

where the L_i are graded B-modules locally free of finite type.

First, M admits a graded epimorphism $L \to M$ with L locally free of finite type: choose an integer N such that M is generated in degrees $\leq N$, and take

$$L = \bigoplus_{i \leq N} M_i \otimes_{\mathcal{O}_S} B(-i),$$

with the morphism (1.2.1). Repeating the construction for successive kernels gives a global graded resolution. After $r+1$ steps, Proposition 1.5 shows locally on affine opens that the final kernel is projective over B, hence locally free as a graded B-module; truncation gives the asserted resolution.

Proposition 1.8. Let S be quasi-compact, let E be locally free of finite type, let $B = \mathrm{Sym}(E)$, and let L be a graded B-module locally free of finite type. Then L admits a finite decreasing filtration by graded submodules $L^{(m)}$ such that

$$\begin{array}{ll} \text{(i)} & L^{(m)} = L \quad \text{for } m \leq m_0, \\ \text{(ii)} & L^{(m)} = 0 \quad \text{for } m \geq m_1, \\ \text{(iii)} & \mathrm{gr}^m L \simeq N_m \otimes_{\mathcal{O}_S} B(m), \end{array}$$

where N_m is locally free of finite type on S.

The question is local on S. Since S is quasi-compact, the degrees of homogeneous bases that occur locally lie between two fixed integers. One argues by induction on the length of this interval. The first step is the canonical map from the lowest non-zero homogeneous component,

$$L_{n_0} \otimes_{\mathcal{O}_S} B(-n_0) \longrightarrow L,$$

which locally identifies the corresponding direct summand. The quotient has local homogeneous bases in higher degrees, and induction applies.

1.9. Generation and the relation

Let F be locally free of finite type on X. By Corollary 1.4 and Proposition 1.7, F comes from a graded module M admitting a finite graded resolution by L_i's. Applying Proj_0 gives an exact sequence of locally free sheaves on X, and therefore in $K^\bullet(X)$

$$\mathrm{cl}(F) = \sum_i (-1)^i \, \mathrm{cl}(\mathrm{Proj}_0(L_i)).$$

Using the filtration of Proposition 1.8, each term is a sum of classes

$$\mathrm{cl}\big(f^* N_{ij}(j)\big) = f^* \, \mathrm{cl}(N_{ij}) \, \xi^j.$$

Thus $K^\bullet(X)$ is generated as a $K^\bullet(S)$-module by the powers of ξ.

For linear dependence, use the canonical epimorphism $f^*E \to \mathcal{O}_X(1)$. Its kernel is locally free of rank r. Hence

$$\lambda^{r+1}(f^*E - \xi) = 0.$$

Since $\lambda_t(-\xi) = 1/(1+\xi t)$, one obtains the relation

$$\sum_{i=0}^{r+1} (-1)^i \, \lambda^i(E) \, \xi^{r+1-i} = 0, \tag{1.9.1}$$

equivalently the characteristic equation of ξ associated with the locally free module E.

1.10–1.11. Linear independence

The proof of the linear-independence assertion is due to J.-P. Jouanolou.

Lemma 1.10. With the hypotheses of Theorem 1.1, let F and G be locally free sheaves of finite type on X, and suppose that they have the same class in $K^\bullet(X)$. Then there is an integer n_0 such that, for all $n \geq n_0$,

(i) $f_*(F(n))$ and $f_*(G(n))$ are locally free of finite type on S,
(ii) $\operatorname{cl} f_*(F(n)) = \operatorname{cl} f_*(G(n))$ in $K^\bullet(S)$.

An equality of classes in $K^\bullet(X)$ is a finite integral combination of relations coming from short exact sequences of locally free sheaves. Twisting by $\mathcal{O}_X(n)$ and applying f_*, for n large enough all higher direct images vanish and all direct images are locally free of finite type; the same relations then hold in $K^\bullet(S)$.

Suppose that

$$f^*(x_0) + f^*(x_1)\xi + \cdots + f^*(x_r)\xi^r = 0. \tag{1.11.1}$$

Writing each x_i as a difference of classes of locally free sheaves and adding suitable summands, (1.11.1) is converted into an equality $\operatorname{cl}(F) = \operatorname{cl}(G)$ for locally free sheaves of finite type on X. By Lemma 1.10, for all sufficiently large n one has

$$\sum_{i=0}^{r} x_i \, \sigma_{n+i}(E) = 0 \quad \text{in } K^\bullet(S), \tag{1.11.2}$$

where $\sigma_p(E)$ denotes the class of $\operatorname{Sym}^p(E)$, and $\sigma_p(E) = 0$ for $p < 0$.

The Koszul complex of the symmetric algebra gives, for $p \geq 1$,

$$\sum_{i=0}^{r+1} (-1)^i \lambda^i(E) \, \sigma_{p-i}(E) = 0. \tag{1.11.3}$$

Using this relation, one descends by induction from large n in (1.11.2). It follows that (1.11.2) holds for all $n \geq -r$. Taking $n = -r$ gives $x_r = 0$; repeating the argument gives successively all $x_i = 0$. This proves the linear independence and hence Theorem 1.1.

Remark 1.12. Let

$$\sigma_t(E) = \sum_{i \geq 0} \sigma_i(E) t^i.$$

The relation (1.11.3) is equivalent to

$$\lambda_t(E) \, \sigma_{-t}(E) = 1.$$

It follows in particular that σ_t is additive and extends to a homomorphism

$$K^\bullet(S) \longrightarrow 1 + K^\bullet(S)[[t]]_+.$$

Remark 1.13. Any sequence of $r+1$ consecutive powers of ξ is a basis of $K^\bullet(X)$ over $K^\bullet(S)$. If

$$x = 1 - \xi^{-1}$$

is the class of a hyperplane section, then $1, x, \ldots, x^r$ is also a basis. The dependence relation becomes

$$\sum_{k=0}^{r+1} \lambda^{r+1-k}(\check{E} - r - 1)\, x^k = 0. \tag{1.13.1}$$

For $E = \mathcal{O}_S^{r+1}$, so that $X = \mathbf{P}^r_S$, this reduces to

$$x^{r+1} = 0.$$

Using relative schemes in Hakim's sense, Theorem 1.1 also makes sense when S is a locally ringed topos; if the final object of the topos is quasi-compact, the proof transports without essential change.

2. Computation of $K^\bullet$ of a projective bundle: the case of perfect complexes

This section is not used in the rest of the Seminar. Here $K^\bullet(X)$ denotes the Grothendieck group of perfect complexes on X.

Theorem 2.1. Let S be quasi-compact, let E be a locally free $\mathcal{O}_S$-module of rank $r+1$, let $X = \mathbf{P}(E)$, and let $\xi \in K^\bullet(X)$ be the class of the complex reduced to $\mathcal{O}_X(1)$ in degree zero. Then $K^\bullet(X)$ is a free $K^\bullet(S)$-module with basis

$$1, \xi, \ldots, \xi^r.$$

2.2. Notation

Let $B = \mathrm{Sym}(E)$. Let $\mathcal{A}$ be the category of $\mathcal{O}_X$-modules, $\mathcal{G}$ the category of graded B-modules, $\mathcal{G}_0$ the full subcategory of graded B-modules locally free of finite type, and $\mathcal{G}'$ the category of $\mathcal{O}_S$-modules. If $L^\bullet$ is a complex on X, put

$$L^\bullet(n) = L^\bullet \otimes_{\mathcal{O}_X} \mathcal{O}_X(n).$$

2.3. The functor Proj_0

We shall need to apply Proj_0 to arbitrary graded B-modules, and not only to quasi-coherent ones. If $Y = \mathrm{Spec}(A)$ and $f : Y \to S$, the isomorphism

$$A \simeq f_*\mathcal{O}_Y$$

gives, by adjunction, a morphism of ringed spaces

$$\widetilde{f} : (Y, \mathcal{O}_Y) \longrightarrow (S, A)$$

and therefore an exact inverse-image functor

$$\widetilde{f}^*(M) = f^{-1}(M) \otimes_{f^{-1}A} \mathcal{O}_Y.$$

For quasi-coherent A-modules this is the usual associated module.

Now let B be graded and quasi-coherent, let M be a graded B-module, and let $X = \mathrm{Proj}(B)$. For an affine open $U \subset S$ and a homogeneous section s of positive degree over U, let $M_{(s),U}$ be the homogeneous component of degree zero of the module of fractions M_s. Thus

$$M_{(s),U} = \big((M|_U) \otimes_{B|_U} (B|_U)_s\big)_0.$$

On the standard affine open

$$D_+(U,s) \simeq \mathrm{Spec}((B|_U)_{(s)})$$

one defines

$$\mathrm{Proj}_0(M)_{U,s} = \widetilde{f}^*_{U,s}(M_{(s),U}).$$

If $V \subset U$ is affine, the modules just defined restrict compatibly. If t is another homogeneous section on U, one uses the standard diagram

$$\begin{tikzcd} D_+(st) \arrow[r, "i"] \arrow[d, "f_{U,st}"'] & D_+(s) \arrow[d, "f_{U,s}"] \\ U \arrow[r, equal] & U \end{tikzcd}$$

and the canonical identifications of the localized degree-zero modules to see that the local sheaves glue. The result is a functor

$$\mathrm{Proj}_0 : \mathcal{G} \longrightarrow \mathcal{A}.$$

It is exact, and for quasi-coherent graded modules it is the usual functor of EGA II. Hence by passage to derived categories it gives

$$\mathrm{Proj}_0 : D^+(\mathcal{G}) \longrightarrow D^+(\mathcal{A}).$$

2.4. The functor F_k

For an $\mathcal{O}_X$-module L and an integer k, set

$$F_k(L) = \bigoplus_{n \geq k} f_*(L(n)). \tag{2.4.1}$$

The functor F_k is left exact and therefore has a right derived functor

$$RF_k : D^+(\mathcal{A}) \longrightarrow D^+(\mathcal{G}). \tag{2.4.2}$$

If L is locally free of finite type, then for k sufficiently large it is F_k-acyclic, by the vanishing theorem used above in 1.3.

For every k there is a functorial morphism

$$\beta : \mathrm{Proj}_0(F_k(F)) \longrightarrow F, \tag{2.5.1}$$

constructed on each $D_+(U,s)$ by the usual localization formula: a fraction represented by a homogeneous section of sufficiently high degree is sent to the corresponding local section of F. In the quasi-coherent case this is the canonical morphism of EGA. Passing to derived categories gives

$$\mathrm{Proj}_0 \circ RF_k \longrightarrow \mathrm{id}_{D^+(\mathcal{A})}. \tag{2.5.2}$$

Proposition 2.6. Let S be quasi-compact, let E be locally free of finite type, let $X = \mathbf{P}(E)$, and let $L^\bullet$ be a perfect complex on X. Then, for k sufficiently large,

(i) $RF_k(L^\bullet)$ is $\mathcal{G}_0$-perfect,

(ii) $\mathrm{Proj}_0\, RF_k(L^\bullet) \longrightarrow L^\bullet$ is an isomorphism in $D^+(\mathcal{A})$.

The assertion is local on S. One may therefore assume S affine. Since $\mathcal{O}_X(1)$ is ample, the perfect complex $L^\bullet$ is locally, and then globally after refining, represented by a bounded complex whose terms are locally free of finite type. For k sufficiently large all those finitely many terms are F_k-acyclic, and the graded modules $F_k(L^i)$ are $\mathcal{G}_0$-perfect. This gives (i), and the morphism (2.5.2) is then an isomorphism term by term, giving (ii).

2.7. Decomposition of a $\mathcal{G}_0$-perfect complex

On $\mathcal{G}$, the functor which associates to a graded B-module its n-th homogeneous component is exact with values in $\mathcal{G}'$; it defines a functor

$$\mathrm{Comp}_n : D^+(\mathcal{G}) \longrightarrow D^+(\mathcal{G}'). \tag{2.7.1}$$

The homomorphism (1.2.1) gives a morphism of functors

$$\mathrm{Comp}_n \otimes_{\mathcal{O}_S} B(-n) \longrightarrow \mathrm{id}_{D^+(\mathcal{G})}\,. \tag{2.7.2}$$

Let $M^\bullet$ be a $\mathcal{G}_0$-perfect complex. We shall show that there exists a finite sequence of distinguished triangles

$$\begin{tikzcd} & P_1^\bullet \arrow[dl] \arrow[dr] & & & P_2^\bullet \arrow[dl] \arrow[dr] & \\ N_1^\bullet & & M^\bullet & N_2^\bullet & & P_1^\bullet \end{tikzcd}$$

$$\cdots \qquad \begin{tikzcd} & P_k^\bullet \arrow[dl] \arrow[dr] & \\ N_k^\bullet & & P_{k-1}^\bullet . \end{tikzcd}$$

where each $P_k^\bullet$ at the last stage is acyclic, the terms $N_i^\bullet$ are of the form

$$N_i^\bullet = K_i^\bullet \otimes_{\mathcal{O}_S} B(d_i),$$

and the $K_i^\bullet$ are perfect complexes on S in the ordinary sense.

Indeed, because $M^\bullet$ is $\mathcal{G}_0$-perfect and S is quasi-compact, there exists a finite covering of S and an integer d_0 such that, on every open of the covering, $M^\bullet$ is represented by a bounded complex of locally free B-modules of finite type, with no homogeneous generators in degrees $< d_0$. Taking d_0 minimal, the morphism (2.7.2) extracts the part of the lowest degree. Its cone has all homogeneous generators in strictly higher degrees. Repeating the process, and using the finiteness of the degree interval from Proposition 1.8, one obtains the desired finite sequence of triangles.

2.8. Proof of Theorem 2.1

Let $L^\bullet$ be a perfect complex on X. For k sufficiently large put

$$M^\bullet = RF_k(L^\bullet).$$

By Proposition 2.6, $M^\bullet$ is $\mathcal{G}_0$-perfect and $\mathrm{Proj}_0(M^\bullet) \simeq L^\bullet$. Applying the decomposition of 2.7 and then Proj_0, one obtains in $K^\bullet(X)$

$$\mathrm{cl}(L^\bullet) = \sum_i \mathrm{cl}\big(f^* K_i^\bullet(d_i)\big) = \sum_i f^*\,\mathrm{cl}(K_i^\bullet)\,\xi^{d_i}.$$

Thus the powers of ξ generate $K^\bullet(X)$ as a $K^\bullet(S)$-module.

The dependence relation is the same relation (1.9.1), now interpreted in the Grothendieck group of perfect complexes. For linear independence one uses the $K^\bullet(S)$-linear homomorphism

$$f_* : K^\bullet(X) \longrightarrow K^\bullet(S).$$

For the structure sheaf and negative twists of the projective bundle one has

$$f_*(1) = 1, \qquad f_*(\xi^n) = 0 \quad (-r \leq n < 0).$$

Multiplying a relation among $1, \xi, \dots, \xi^r$ by suitable negative powers of ξ and applying f_* successively gives all its coefficients equal to zero. This completes the proof of Theorem 2.1.